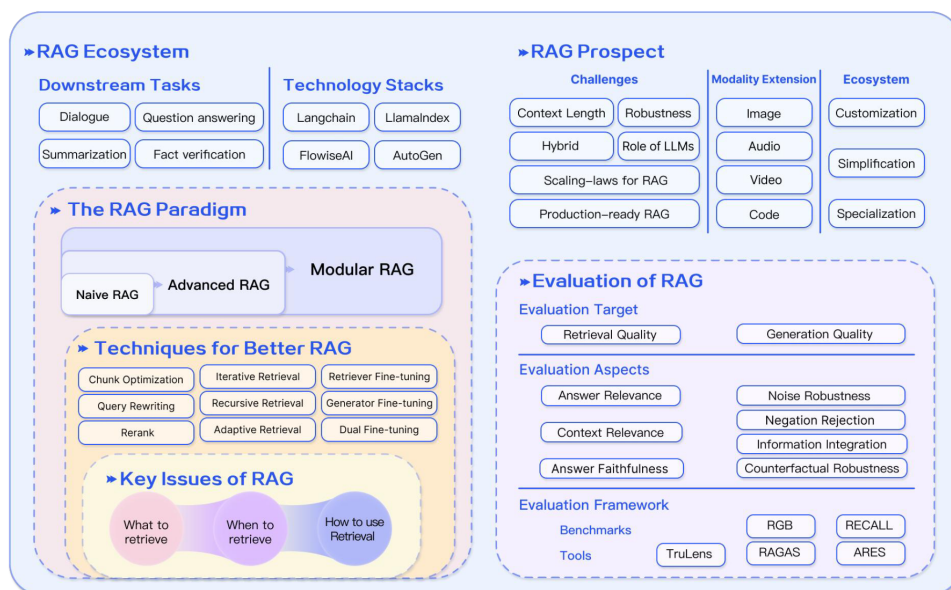


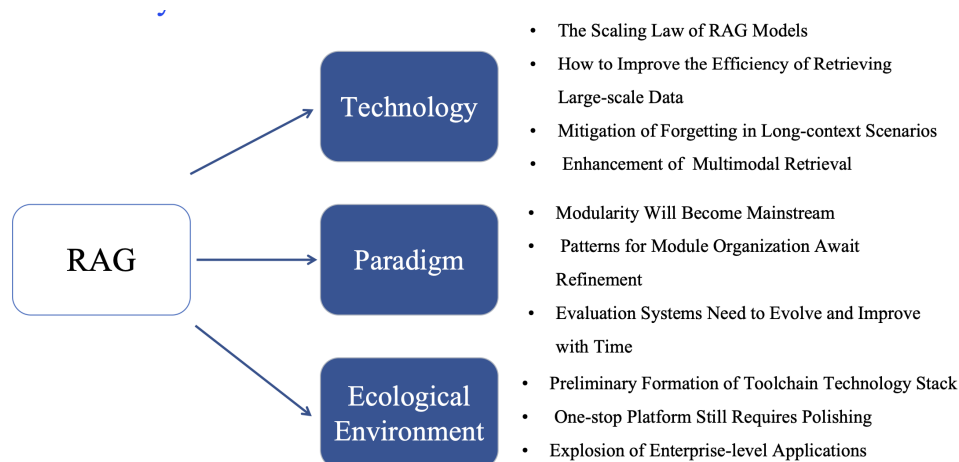
RAG技术调研文档

- RAG

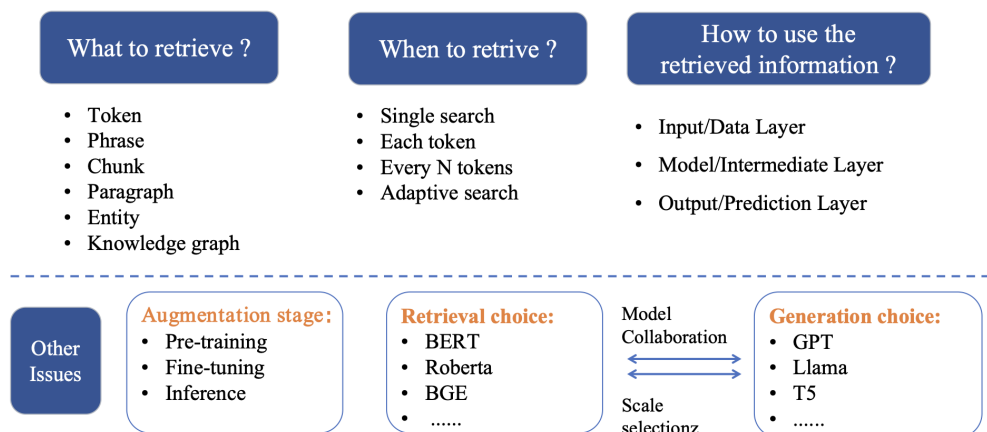


○

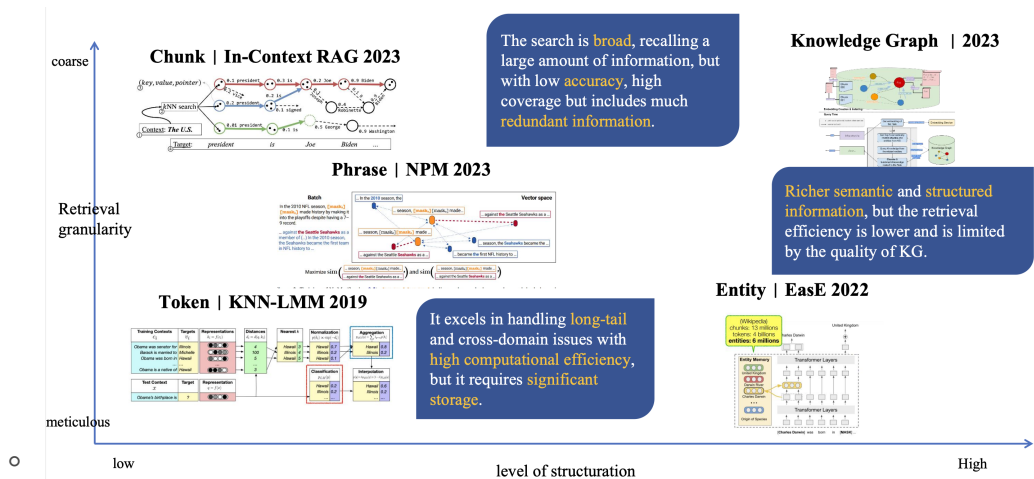
- 关键点



- 构建RAG的核心问题（三个）

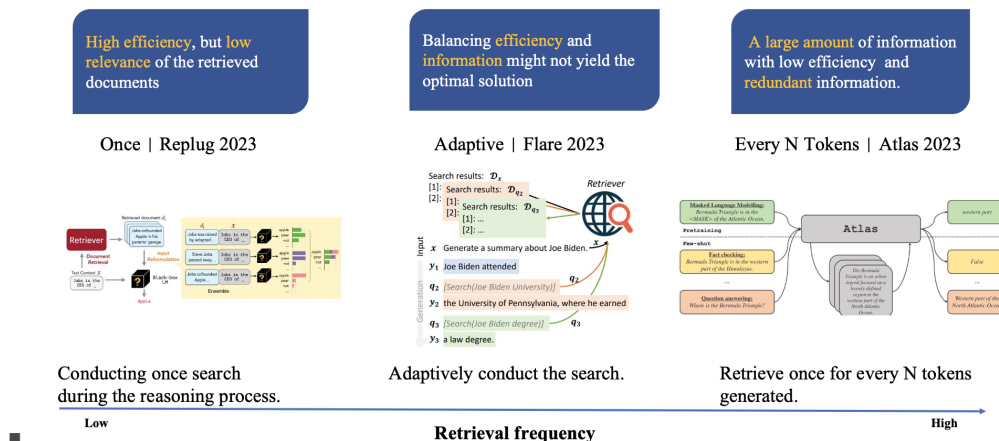
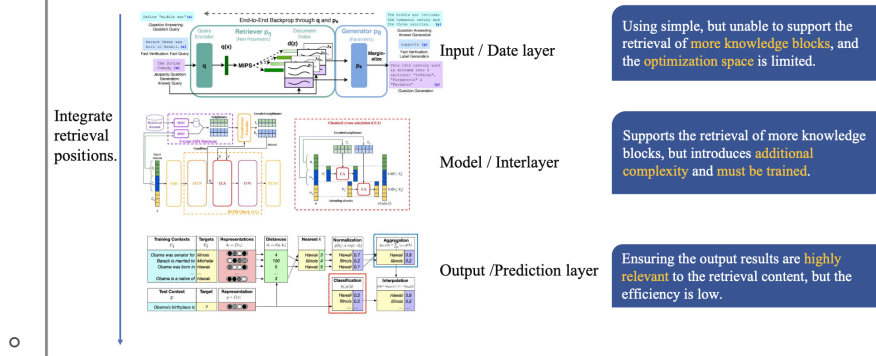


○



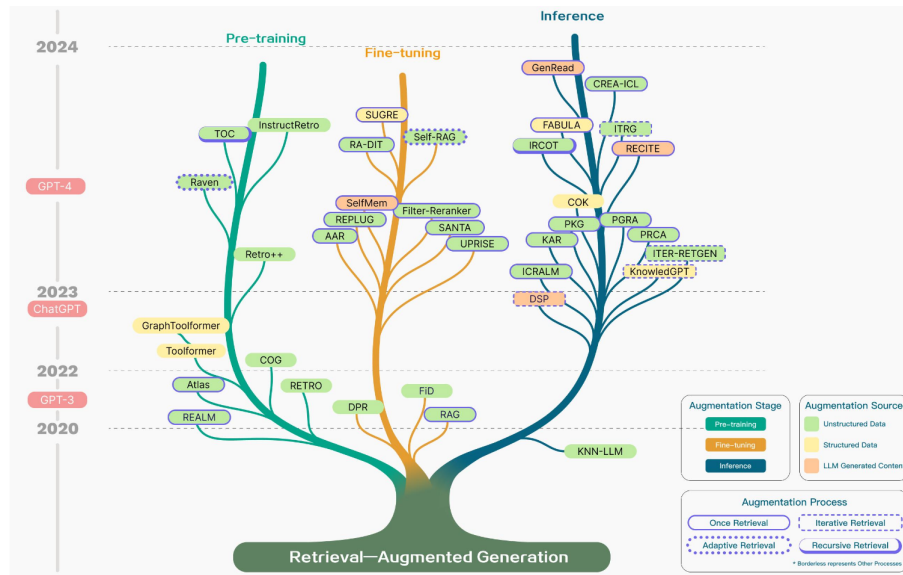
- Tokens
 - 论文: [1911.00172 \(arxiv.org\)](https://arxiv.org/abs/1911.00172)
 - 解说: [香依读 | 用上文K最近邻特征表示增强语言模型 – 知乎 \(zhihu.com\)](https://www.zhihu.com/question/35484444)
- Chunk
 - 论文: [2302.00083 \(arxiv.org\)](https://arxiv.org/abs/2302.00083)
 - 解说: [TACL | In-Context Retrieval-Augmented Language Models – 知乎 \(zhihu.com\)](https://www.zhihu.com/question/68444444)
 - code: <https://github.com/AI21Labs/in-context-ralm>
- Phrase
 - 论文: [2212.01349 \(arxiv.org\)](https://arxiv.org/abs/2212.01349)
 - code: [code](#)
 - 解说: [Meta发布首个「非参数化」掩码语言模型NPM：吊打500倍参数数量的GPT-3 – 知乎 \(zhihu.com\)](https://www.zhihu.com/question/68444444)

Integrating the retrieved information into different layers of the generation model,during inference process.

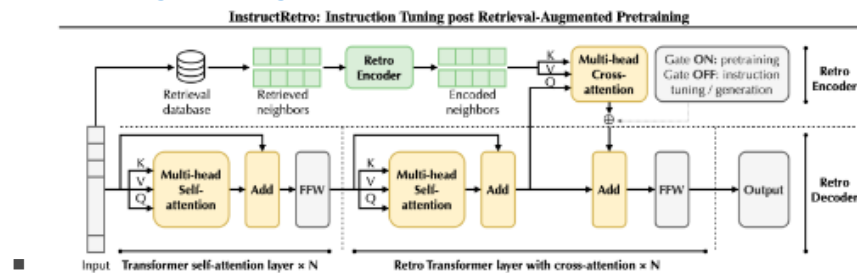


- **Retrieval frequency**
 - Once
 - paper: [2301.12652 \(arxiv.org\)](https://arxiv.org/abs/2301.12652)

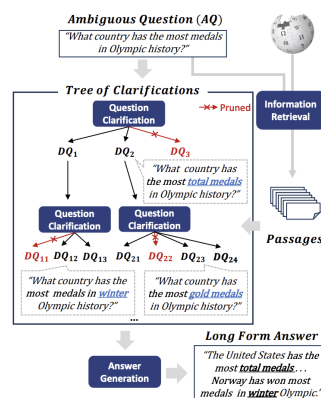
- 解说: [REPLUG: 检索增强的黑盒子语言模型 – 知乎 \(zhihu.com\)](#)
- Adaptive
 - paper: [2305.06983 \(arxiv.org\)](#)
 - code: [FLARE](#)
 - 解说: [使用主动检索增强生成 FLARE 实现更优越的 RAG – 知乎 \(zhihu.com\)](#)
- Every N tokens
 - paper: [2208.03299 \(arxiv.org\)](#)
 - 解说: [code](#)
- 主要发展



- 根据阶段划分
 - 预训练
 - InstructRetro: [InstructRetro: Instruction Tuning post Retrieval-Augmented Pretraining \(arxiv.org\)](#)

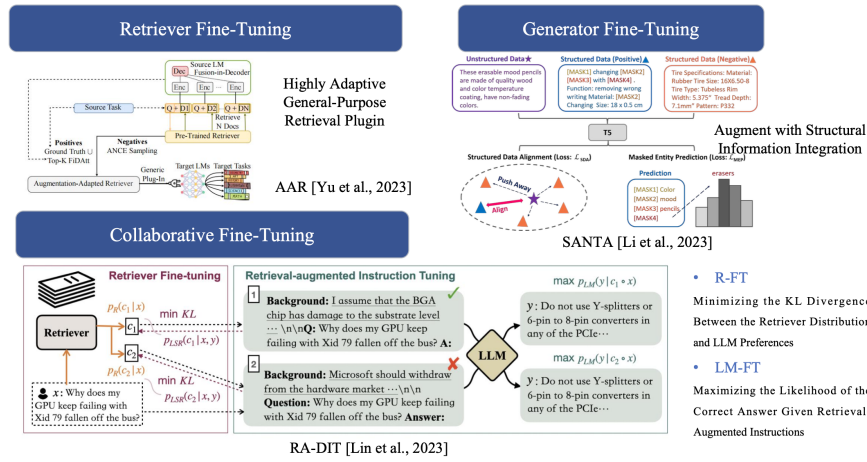


- code: [Megatron-LM/tools/retro at InstructRetro · NVIDIA/Megatron-LM · GitHub](#)
- TOC: [2310.14696 \(arxiv.org\)](#)



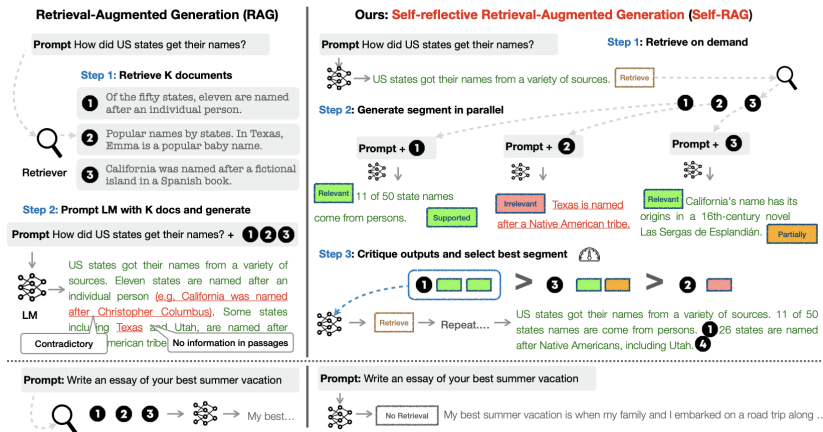
- code: [TOC](#)

□ 微调



○ RA-DIT [Lin et al., 2023]

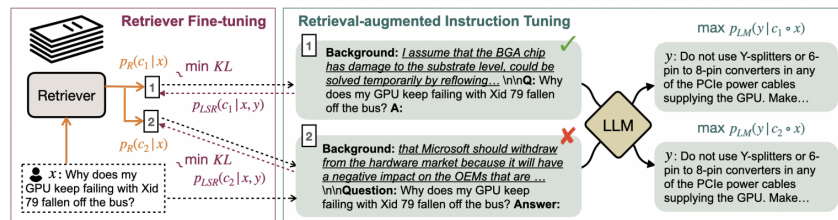
○ Self-RAG: [2310.11511 \(arxiv.org\)](https://arxiv.org/abs/2310.11511)



■ 解说: SELF-RAG: 自我反思检索增强生成, 提升语言模型生成质量与准确性 – 知乎 (zhihu.com)

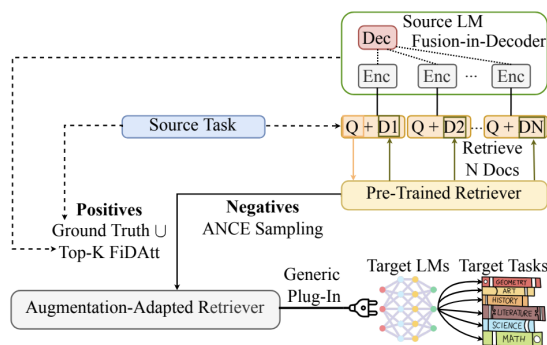
■ code: [self-rag](#)

○ RA-DIT: [2310.01352 \(arxiv.org\)](https://arxiv.org/abs/2310.01352)



■ 解说: RA-DIT: Retrieval-Augmented Dual Instruction Tuning – 知乎 (zhihu.com)

○ AAR: [2305.17331 \(arxiv.org\)](https://arxiv.org/abs/2305.17331)

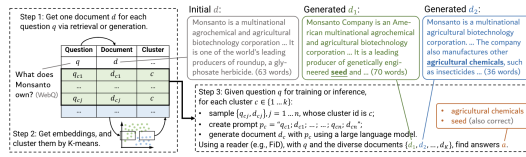


■ code: [code](#)

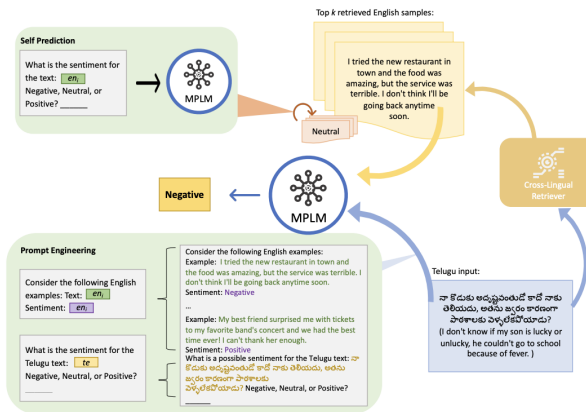
○ SANTA: [2305.19912 \(arxiv.org\)](https://arxiv.org/abs/2305.19912)

■ code: <https://github.com/OpenMatch/OpenMatch>

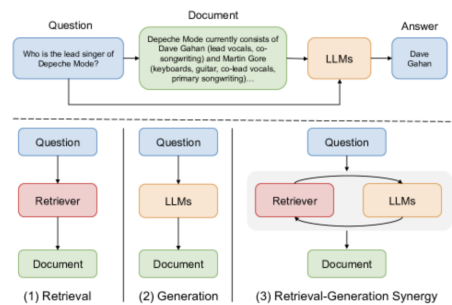
- Self-Mem: [2305.02437 \(arxiv.org\)](https://arxiv.org/abs/2305.02437)
 - code: <https://github.com/Hannibal046/SelfMemory>
 - 解说: [腾讯和北大提出 | 生成具有自我记忆的检索增强文本 - 知乎 \(zhihu.com\)](https://zhuanli.zhihu.com/question/67411111)
- 推理
 - GenRead: [2209.10063 \(arxiv.org\)](https://arxiv.org/abs/2209.10063)



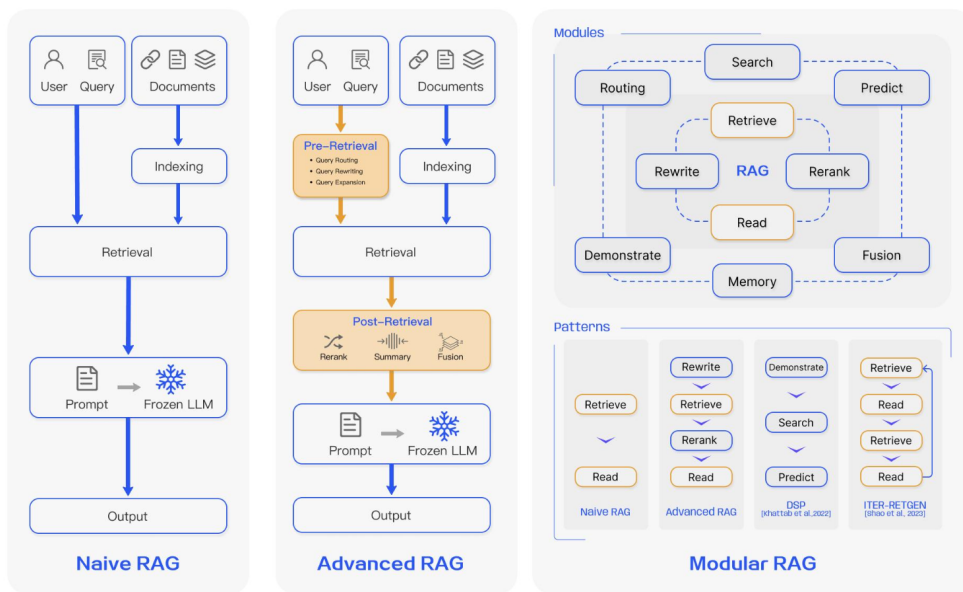
- 泛读: [泛读](https://arxiv.org/abs/2311.06595)
- CREA-ICL: [2311.06595 \(arxiv.org\)](https://arxiv.org/abs/2311.06595)



- ITRG: [ITRG](https://arxiv.org/abs/2311.06595)



- 根据检索数据类型划分
 - 非结构化数据 (文本)
 - 半结构化数据 (PDF)
 - 结构化数据 (知识图谱、三元组)
 - LLM生成的数据
- RAG技术总览



o

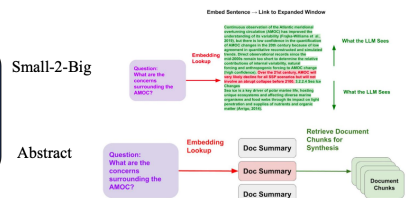
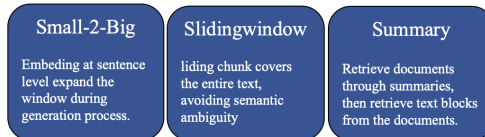
方法1-Naive RAG（静态）-直接使用静态题库

- 核心：Read-Retrieve-Generate
- 三步：建立索引-链接-生成
 - 将文档/文本切分成chunk-使用编码器编码-建立数据库
 - 从数据库中取出与prompt最相似的k个向量
 - 将抽取的向量与原始的prompt进行拼接输入模型

方法2-Advanced RAG（静态）-加入了pre-retrieval和post-retrieval

- 核心：Index优化-检索前处理-检索-检索后处理-生成
- 建立索引部分的优化
 - 滑动窗口、细粒度的切分（词级别/句子级别）、加入原始信息（页码、文件名、作者、文件类型、时间戳等信息）

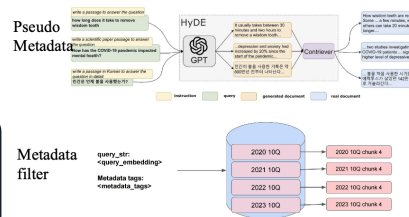
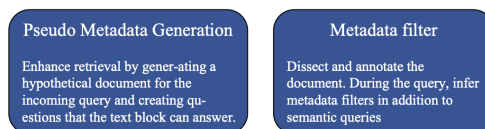
Chunk Optimization



Adding Metadata

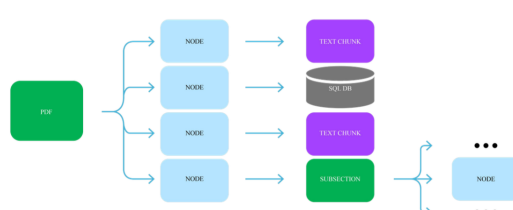
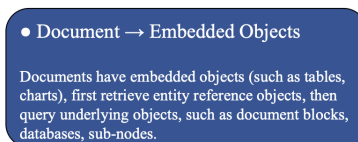
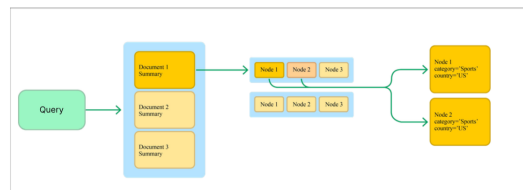
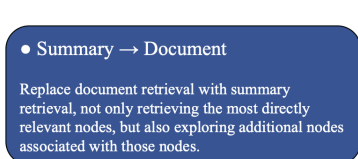


Metadata Filtering/Enrichment



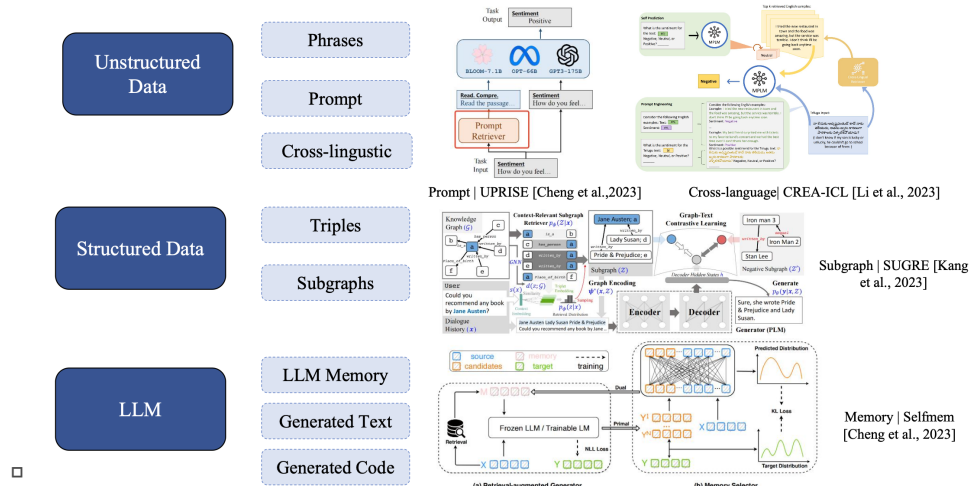
□

Hierarchical Organization of Retrieval Corpora



□

- pre-retrieval处理 (主要处理检索前的query相关的内容)
 - 链接链路 (retrieve routes)、总结、重写、置信度评估 (confidence judgement)



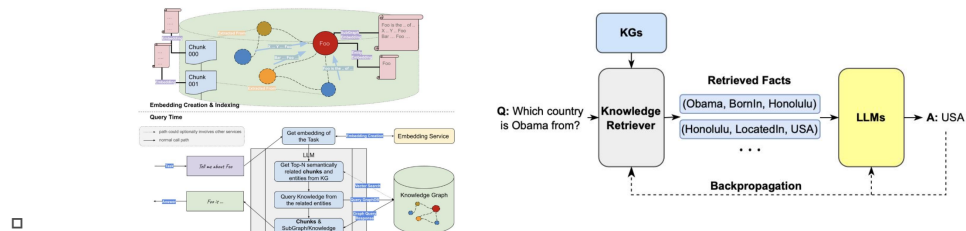
- UPRISE: [paper](#)
- CREA-ICL: [2311.06595 \(arxiv.org\)](#)
- SURGE: [2305.18846 \(arxiv.org\)](#)
- Self-Mem: [2305.02437 \(arxiv.org\)](#)

➤ GraphRAG

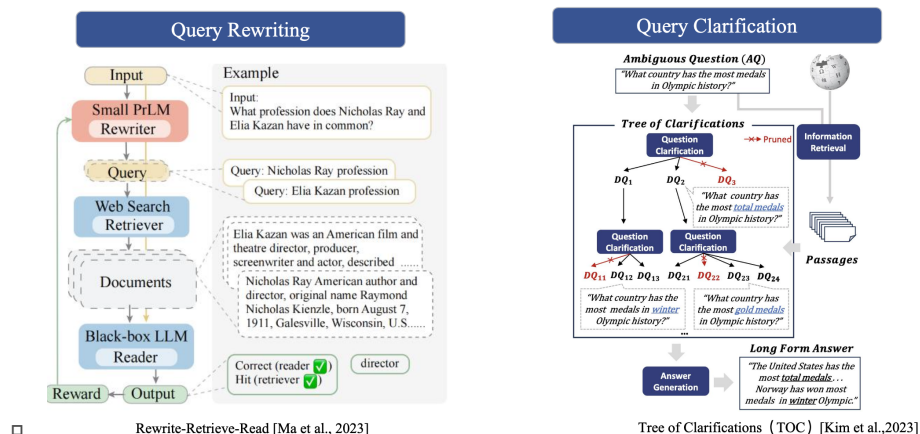
- Extract entities from the user's input query, then construct a subgraph to form context, and finally feed it into the large model for generation.

➤ Implementation

- Use LLM (or other models) to extract key entities from the question.
- Retrieve subgraphs based on entities, delving to a certain depth, such as 2 hops or even more.
- Utilize the obtained context to generate answers through LLM.

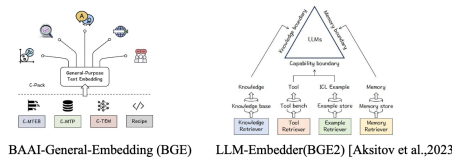


Questions and answers do not always possess high semantic similarity; adjusting the Query can yield better retrieval results.

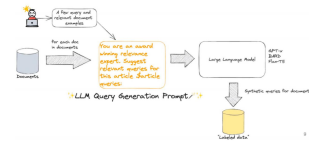


- 优化embedding

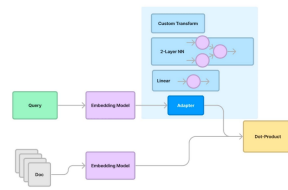
Selecting a More Suitable Embedding Provider



Fine-tuning the Embedding Model



Fine-tuning According to Domain-Specific Repositories and Downstream Tasks

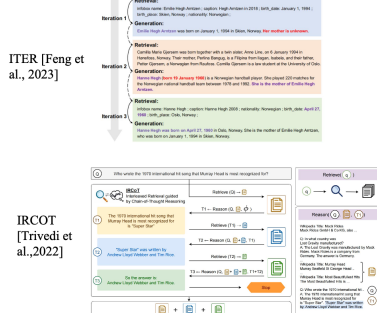


Fine-tuning the Adapter Module to Align the Embedding Model with the Retrieval Repository

优化 retrieval

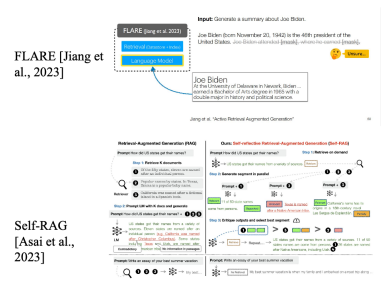
Iterative

Iteratively Retrieving from the Corpus to Acquire More Detailed and In-depth Knowledge



Adaptive

Dynamically Determined by the LLM, the Timing and Scope of Retrieval

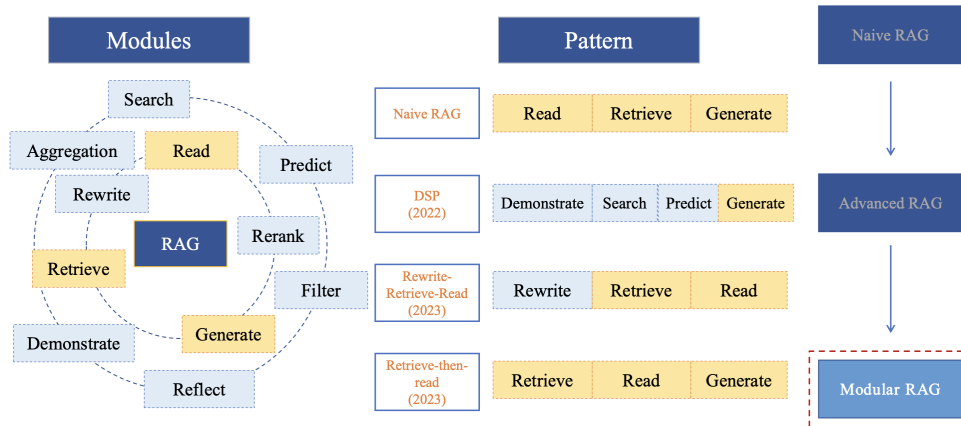


post-retrieval处理 (主要处理检索后的输入LLM前的内容)

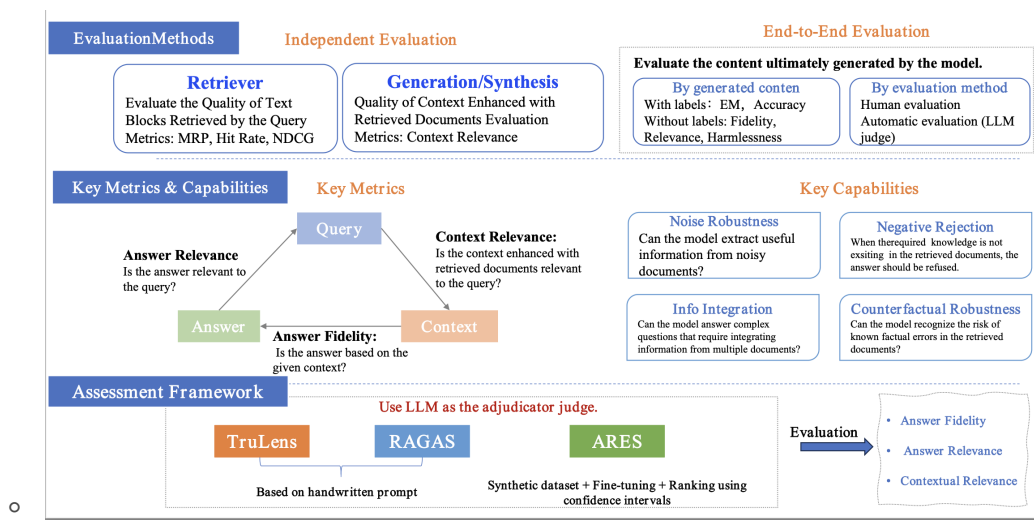
乱序、过滤内容

方法3-Modular-RAG (动态)

Modular RAG



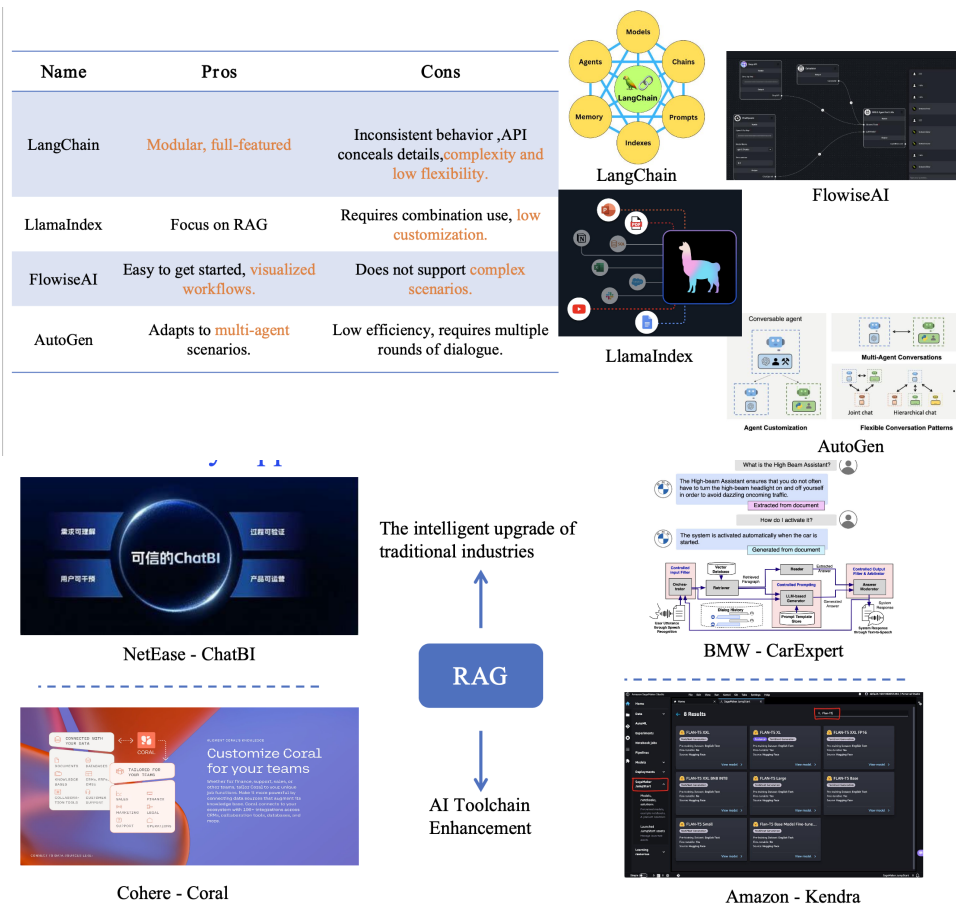
评估



- 相关文章：
 - 应用论文
 - 常识推理
 - 论文：2210.12887 (arxiv.org)
 - 解说：Retrieval Augmentation for Commonsense Reasoning: A Unified Approach – 知乎 (zhihu.com)
 - Fact Checking
 - 论文–RAG–2020：2005.11401 (arxiv.org)
 - 解说：NeurIPS 2020|RAG：为知识密集型任务而生 – 知乎 (zhihu.com)
 - 论文：2022.naacl-main.162.pdf (aclanthology.org)
 - code：code
 - Dialog
 - 论文：arxiv.org/pdf/2208.03188
 - 解说：BlenderBot3: 基于知识的对话机器人 – 知乎 (zhihu.com)
 - 论文：2107.07566 (arxiv.org)
 - 解说：ACL–2022|Internet–Augmented Dialogue Generation – 知乎 (zhihu.com)
 - Summary
 - 论文：2305.06983 (arxiv.org)
 - 解说：Active RAG -- FLARE 详解 – 知乎 (zhihu.com)
 - Machine Translation
 - 论文：2010.00710 (arxiv.org)
 - 解说：MT论文速读：使用kNN算法做机器翻译 – 知乎 (zhihu.com)
 - 论文：2205.12674 (arxiv.org)
 - 解说：[ACL 2023 Tutorial Retrieval-based LMs]: Applications梳理总结 – 知乎 (zhihu.com)
 - code：code
 - Code Generation
 - 论文：2207.05987 (arxiv.org)
 - 解说：「论文阅读笔记」DOCPROMPTING: GENERATING CODE BY RETRIEVING THE DOCS – 知乎 (zhihu.com)
 - 论文：2205.12910 (arxiv.org)
 - code：code
 - Natural Language Inference & Sentiment Analysis

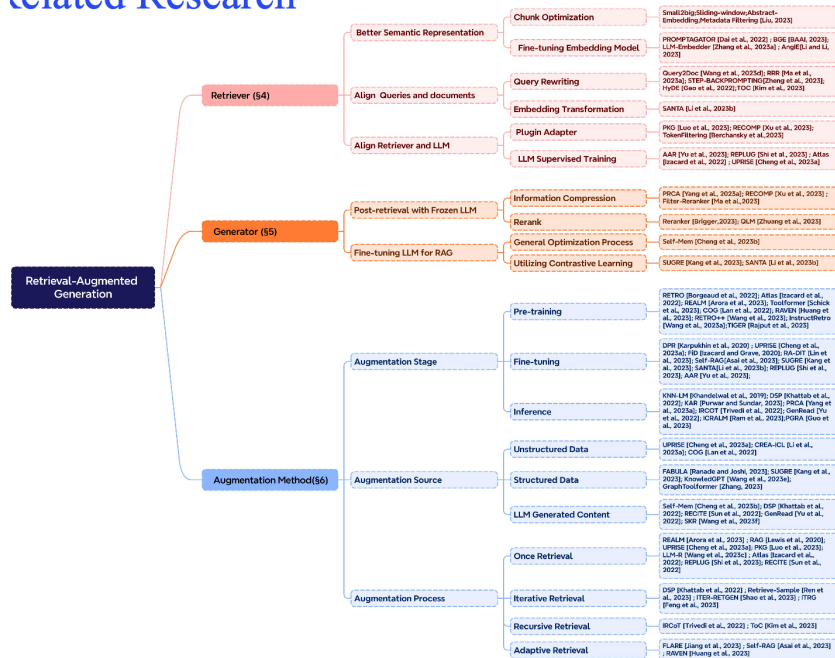
- 论文: arxiv.org/pdf/2303.13824
 - 解说: 【ICLR23】大模型上下文学习的无限扩展方法: kNN Prompting – 知乎 (zhihu.com)
- 论文: [2212.01349](https://arxiv.org/abs/2212.01349) (arxiv.org)
 - code: [code](#)
 - 解说: Meta发布首个「非参数化」掩码语言模型NPM: 吊打500倍参数量的GP T-3 – 知乎 (zhihu.com)
- Q&A
 - 论文: [2112.04426](https://arxiv.org/abs/2112.04426) (arxiv.org)
 - 解说: 以小25倍参数量媲美GPT-3的检索增强自回归语言模型: RETRO – 知乎 (zhihu.com)
 - 论文: [2002.08909](https://arxiv.org/abs/2002.08909) (arxiv.org)
 - 解说: 《REALM: 检索增强的预训练语言模型》阅读笔记 – 知乎 (zhihu.com)
 - 论文: [2208.03299](https://arxiv.org/abs/2208.03299) (arxiv.org)
 - 解说: 文本生成系列之retrieval augmentation (进阶篇Atlas) – 知乎 (zhihu.com)

工业落地



RAG综述:

Related Research



■

■ 论文: <https://arxiv.org/abs/2312.10997>

■ 仓库: <https://github.com/Tongji-KGLLM/RAG-Survey>

■ 知乎概述:

- [ACL 2023 Tutorial Retrieval-based LMs]: Applications 梳理总结 – 知乎 (zhihu.com)
- 深度好文! 最全的大模型 RAG 技术概览 – 知乎 (zhihu.com)

○ 问题&挑战

Further address the challenges faced by RAG itself

Long context

- Retrieved content is excessive, **exceeding window limit**.
- The context is too long to result **Lost in the Middle**.
- If the context **window is not limited**, is there still a need for RAG?

Coordination with FT

- How to simultaneously leverage the effects of **RAG** and **FT**.
- How do the two coordinate, how are they organized, is it in **Pipeline**, **alternating**, or **end-to-end**?

The role of LLMs

- LLM can be used for **retrieval** (LLM generation replaces retrieval, retrieving from LLM memory), for **generation**, and for **evaluation**. How to further explore the **potential** of LLM in RAG.

Robustness

- How to handle the **incorrect** content retrieved.
- How to **filter** and **verify** the content retrieved.
- How to improve the model's **resistance to toxicity and noise**

Scaling Law

- Does the RAG model satisfy the **Scaling Law**
- Does RAG exhibit, or under what scenarios does it exhibit an **Inverse Scaling Law**

Engineering Practice

- How to reduce the **latency** of retrieving ultra-large-scale corpora.
- How to ensure that the content retrieved is not **leaked** by large models

■

● 相关开源工具

- <https://github.com/weaviate/Verba>
- https://github.com/BaranziniLab/KG_RAG
- [easy-rag](#)
- 参考落地网站
- <https://aws.amazon.com/cn/kendra/>